

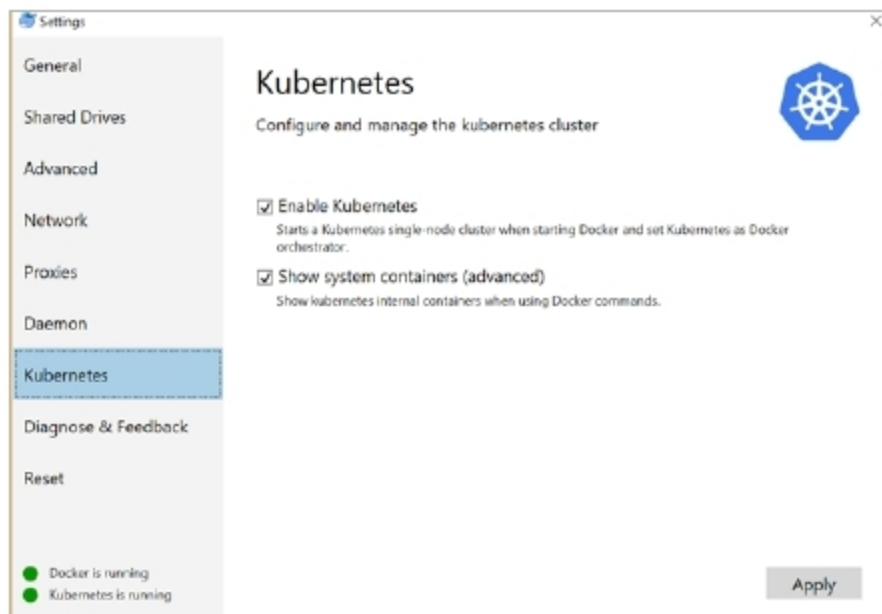


Figure 1: Enabling Kubernetes in Docker for Windows

Common configuration

Creating the default network: To configure the Docker Engine so that a default NAT network is not created, use the following code:

```
{
  "bridge" : "none"
}
```

Setting up the Docker security group: When logged into the Docker host and running Docker commands locally, these commands are run through a named pipe. By default, only members of the administrators' group can access the Docker Engine through the named pipe. To specify a security group that has this access, use the group flag, as shown below:

```
{
  "group" : "docker"
}
```

Proxy configuration

To set proxy information for Docker search and Docker pull, create a Windows environment variable named `HTTP_PROXY` or `HTTPS_PROXY` and a value of the proxy information. This can be completed with PowerShell using the following command and restarting the Docker service.

```
[Environment]::SetEnvironmentVariable("HTTP_PROXY", "http://
username:password@proxy:port/", [EnvironmentVariableTarget]::
Machine)
Restart-Service docker
```

Prerequisites for the Kubernetes cluster

The Docker desktop for Windows 18.06.0-ce-win70 CE includes a standalone Kubernetes server and client, as well as Docker CLI integration. The Kubernetes server that runs locally within your Docker instance is not configurable. It is a single node cluster and is only for local testing.

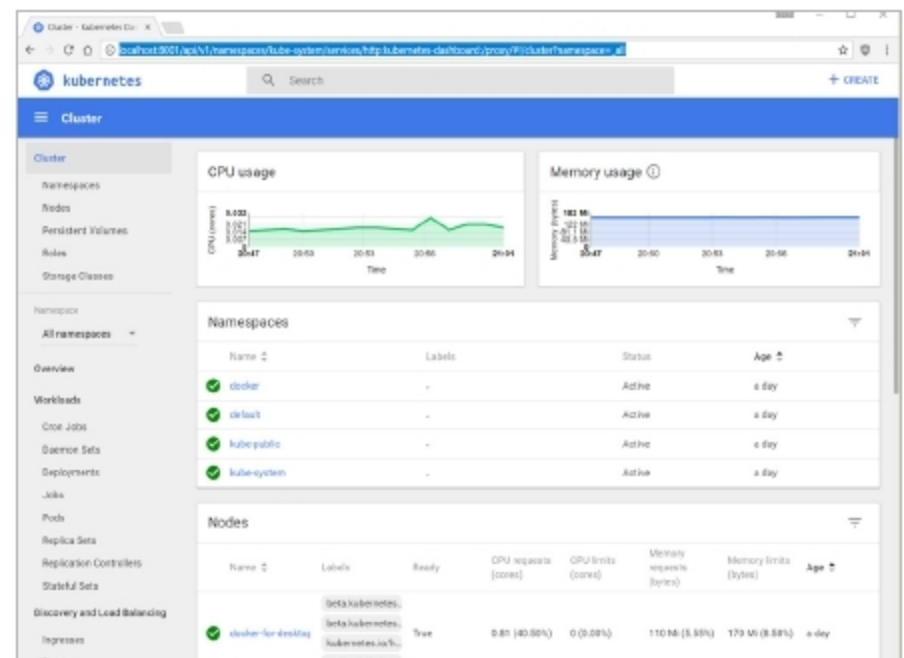


Figure 2: Kubernetes dashboard

Setting up Kubernetes

It's easy to get started; just click 'Enable Kubernetes', after which Docker for Windows will download and start the images you need. I also check 'Show system containers' because I like to see what's hidden from me, but you can decide that for yourself.

This will also install *kubectl* — the command line client for operating your Kubernetes cluster.

By default, you won't get the Kubernetes dashboard, so use the following commands to set it up.

```
kubectl apply -f https://raw.githubusercontent.com/
kubernetes/dashboard/master/src/deploy/alternative/
kubernetes-dashboard.yaml
```

Before you access the dashboard, run the following command:

```
kubectl proxy
```

...and then visit `http://localhost:8001/ui`.

Now you can run through all the cool Kubernetes tutorials from the convenience of your Windows 10 machine. Multi-node clusters for Kubernetes on Windows are still in beta and I am also evaluating them. Look out for updates on these in my coming articles. **END** 🐧

Reference

[1] <https://bit.ly/2SWfu7R>

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